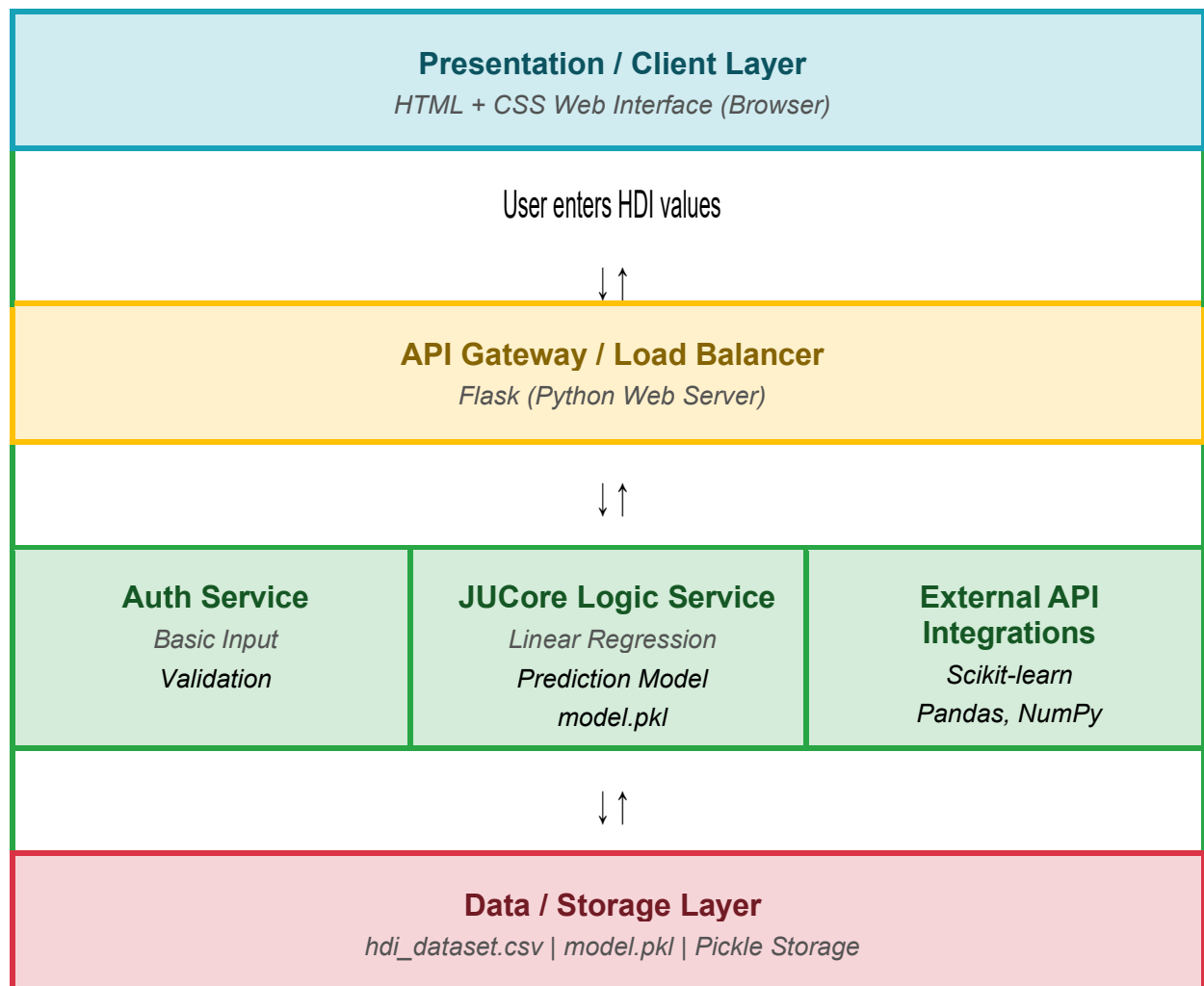


Solution Architecture

Date	4 July 2026
Team ID	xxxxxxx
Project Name	Human Development Index (HDI) Prediction Using Machine Learning
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides the user interface where users enter Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI values and view the prediction result.</i>	<i>HTML, CSS</i>
<i>API Gateway</i>	<i>Receives user requests, processes the input, communicates with the machine learning model, and returns the prediction result.</i>	<i>Flask, Python</i>
<i>Microservices</i>	<i>Performs data validation, loads the trained Linear Regression model, predicts the HDI score, and determines the HDI category.</i>	<i>Scikit-learn, Pandas, NumPy, Pickle</i>
<i>Database</i>	<i>Stores the HDI dataset used for training and the saved machine learning model used for prediction.</i>	<i>hdi_dataset.csv, model.pkl</i>